MUSD PEG DESK · MEZO BUILDATHON WAVE 1

MUSD's peg defense has no front door.

Redemption, burning 1 MUSD for $1 of BTC, is what holds MUSD at $1. People use it every day, yet there is no product for it: redeemers fill a raw form, and borrowers never see it coming.

892.5K

MUSD redeemed on mainnet in 12 days: 500 `Redemption` events, Oct 8–20, 2026.

6 fields

The documented way to redeem: hand-filled hints in the block explorer. Most hand-picked sizes revert because the last loan must keep 1,800 MUSD of debt.

0 tools

tell a borrower that their loan is next in the redemption queue, or what a depeg would take from them.

MUSD Peg Desk: one page that serves **both sides** of redemption, read live from Mezo, with no backend.

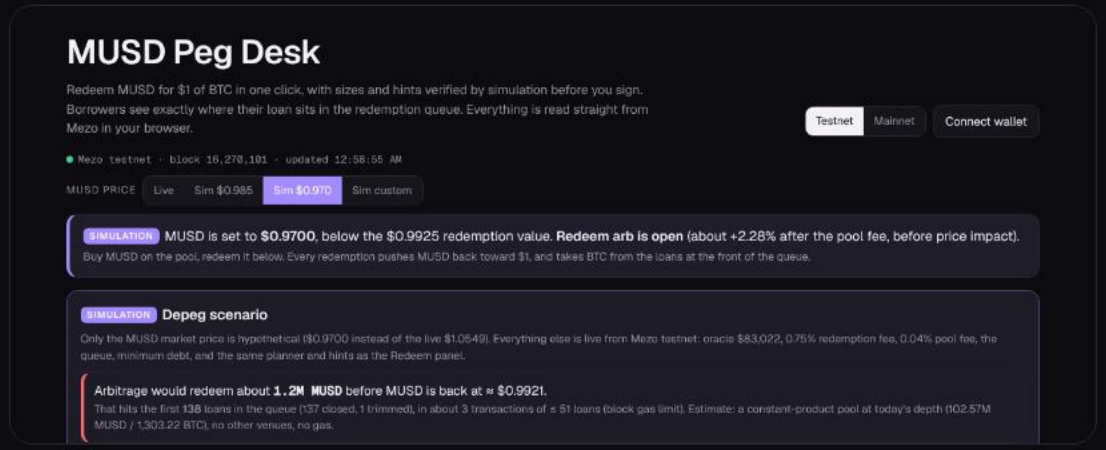
Sources: mainnet TroveManager Redemption logs (explorer API); mezo.org/docs, MUSD Redemptions

WHO IT SERVES

Redeemers keep the peg. Borrowers pay for it. Peg Desk serves both.

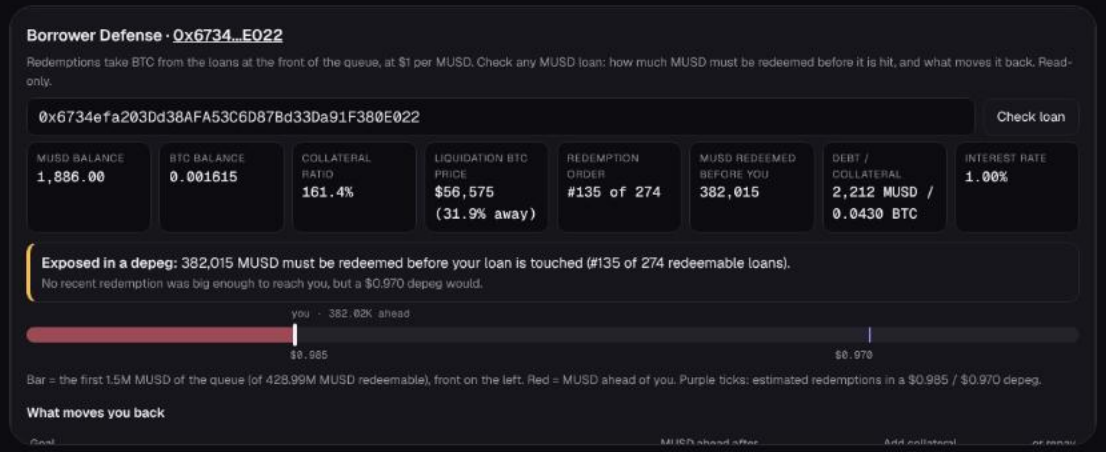
Redeemers, keepers and MUSD holders

- Live peg signal: redeem arb or mint-and-sell, after fees
- Only sizes that fill on-chain; one click to send
- SIMULATION Depeg presets ($0.985 / $0.970 / custom): profit per size, using the real fee and queue math



MUSD borrowers: Borrower Defense

- Your place in the queue and the **MUSD that must be redeemed before you are hit**
- Would the largest recent redemption, or a $0.985 / $0.970 depeg, reach you?
- Exactly how much BTC to add, or MUSD to repay, to move back



Screens: live Mezo testnet, 274 loans

HOW IT WORKS

It mirrors `redeemCollateral` step by step, then proves it with `eth_call`.

01
Read
Every loan, ratio, fee and price in a few Multicall3 batches

02
Plan
Walk SortedTroves like the contract: close loans in full, trim to minNetDebt, cap to the block gas limit

03
Hints
Centred NICR, findInsertPosition, HintHelpers cross-check

04
Guards
TCR ≥ 110%, max-fee cap (+ pending governance rate), balance

05
Simulate
eth_call + estimateGas from your wallet or a real holder

06
Sign
EIP-1193 wallet on chain 31611; Redemption event decoded



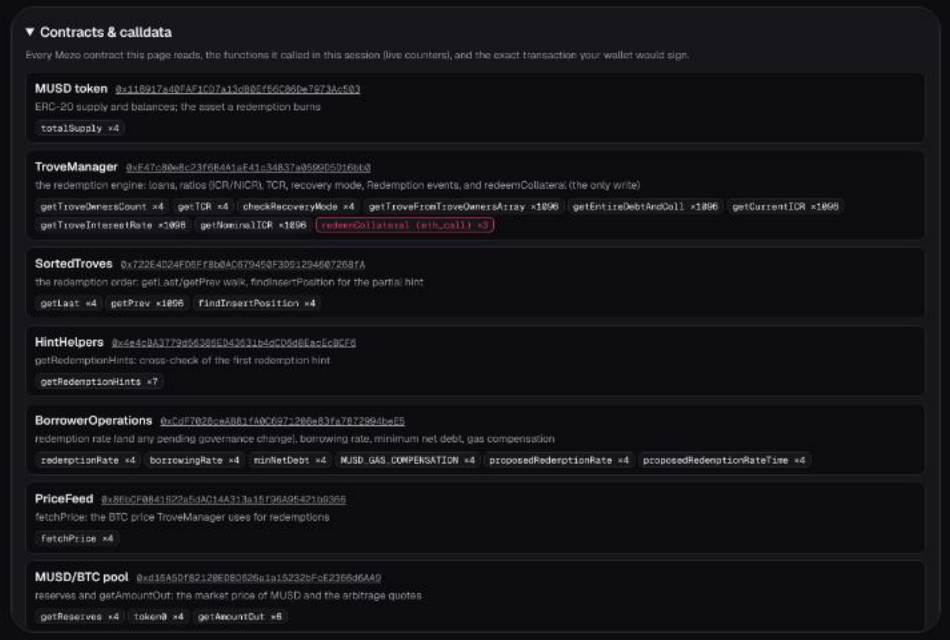
Why it matters

- **"Full" sizes** close whole loans, so they never depend on the BTC price
- Every size is verified by simulation before the wallet opens
- `redeemCollateral` has no fee cap, so Peg Desk adds one on the client
- A 51-loan redemption is planned within the 10M gas block limit and simulated successfully (8.2M gas)

MEZO INTEGRATION DEPTH

7 Mezo contracts read live. One write: the peg mechanism itself.

CONTRACT	CALLS
TroveManager	`getEntireDebtAndColl` · `getCurrentICR` · `getNominalICR` · `getTCR` · `checkRecoveryMode` · Redemption events · `redeemCollateral (write)`
SortedTroves	`getLast` · `getPrev` (queue walk) · `findInsertPosition`
HintHelpers	`getRedemptionHints` (first-hint cross-check)
BorrowerOperations	`redemptionRate` · `proposedRedemptionRate` · `minNetDebt` · `borrowingRate`
PriceFeed	`fetchPrice`
MUSD	`totalSupply` · `balanceOf`
MUSD/BTC pool	`getReserves` · `getAmountOut` · factory `getFee`



Composable by design: it joins the MUSD/BTC pool price to TroveManager redemptions. In Wave 2 the executor makes "swap + redeem" one atomic transaction.

Testnet + mainnet addresses in the README; live call counters in the app's “Contracts & calldata” drawer

BUSINESS PATH · WAVE 2 ROADMAP

Earn where the peg is defended.

Keeper fee

10% of realized profit on redemptions routed through the executor, charged only when min-out clears. Mainnet flow is about 2.2M MUSD a month; at a 0.5–1% edge, a 25% routing share is about **$280–560 a month**, and it grows with every depeg.

Alerts

Borrower subscription: Telegram or web-push when a watched loan enters the first N MUSD of the queue, or when a depeg scenario would reach it. Target: 2–5 MUSD per loan per month. The page stays free.

Wave 2 · Nov 2–15

- Atomic arb executor (swap → redeemCollateral, min-out), verified on Mezo testnet
- Borrower alerts bot

December

- Mainnet sends after external review
- Keeper bot at threshold
- Paid alerts live

Q1 2027

- SDK + embeddable widget: "queue position" and "redeem at $1" inside wallets and Mezo front-ends

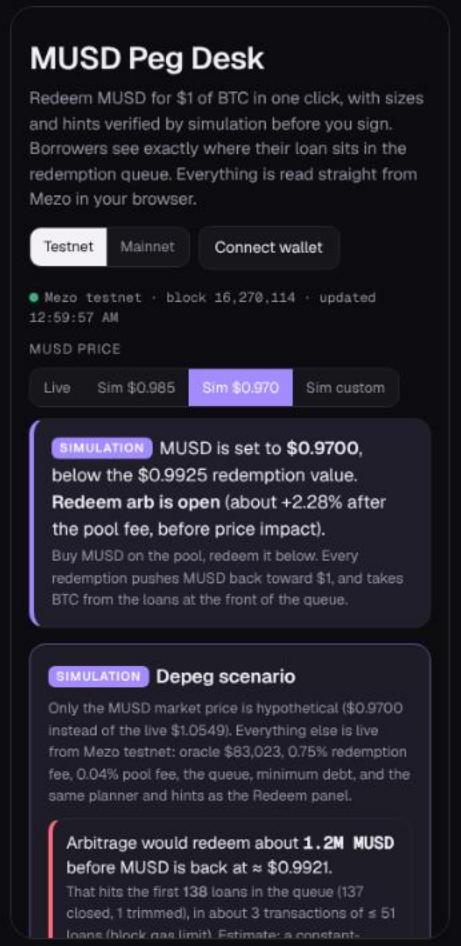
All revenue in MUSD; it extends MUSD utility rather than detouring from it

PROOF · TRY IT IN 60 SECONDS

It works on Mezo testnet today.

Live app	LIVE_URL
Code (MIT)	REPO_URL
Video	VIDEO_URL

- `tools/verify.py` (Playwright, live chain): `10 / 10 checks pass`: dry-run, pre-sign checks, contracts drawer, SIMULATION labels, Borrower Defense, max-fee guard, wallet payload replayed as eth_call, 390 px, 0 console errors
- Full-size dry-run of 2,012 MUSD: `passed`, about 553K gas. A 51-loan, 134K MUSD redemption: `passed`, 8.2M gas
- 274 testnet loans read in exact SortedTroves order; mainnet read-only view live
- Try it: click a `full` chip → Dry-run → Sim $0.970 → “exposed in a $0.970 depeg”



Team: NAME, builder (X: X_HANDLE). Independent open-source tool, not affiliated with Mezo.